Name _____ Period_____

String Operations

Your Tasks (Mark these off as you go)

- ☐ Trace concatenation, length, and case conversions
- ☐ Retrieve substrings using indexes
- ☐ Use escape sequences to format output
- ☐ Write an algorithm to rotate a String
- ☐ Receive credit for this lab guide

☐ Trace concatenation, length, and case conversions

Trace the code below. Write the exact five lines that will be printed to the console. Remember that spaces count as characters.

```
String first = "Java";
String second = "Strings";
String label = first + " " + second;

System.out.println(label);
System.out.println(label.length());
System.out.println(label.toUpperCase());
System.out.println(second.toLowerCase());

label.toUpperCase();
System.out.println(label);
```


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□ Retrieve portions of a String using indexes

Refer to the code segment below,

```
String word = "Trailblazer";
Character: T r a i l b l a z e r
Index:     0 1 2 3 4 5 6 7 8 9 10
```

Indicate the output for each of the following

```
System.out.println(word.substring(5));
```

```
System.out.println(word.substring(0, 5));
```

```
System.out.println(word.substring(2, 7));
```

□ Use escape sequences to format output

Write Java code that produces the display below.

```
Student:   "Ada"
Folder:    C:\Java\Labs
Status:    Ready!
```

Use one String variable and one System.out.println statement. Your code must use each of these escape sequences at least once: \n, \t, \", and \\.

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☐ Write an algorithm to rotate a String

Complete the expressions to complete the algorithm needed to divide Bicycle into two halves. Scramble the word by swapping the first and second halves. Then, convert the result to lower case

```
String word = "Bicycle";
```

1. int splitPosition = _____;
2. String firstHalf = _____;
3. String secondHalf = _____;
4. String rotated = _____;

What will rotated contain?

Consider a word with an odd number of characters, like “Awesome”. What would be the output if you applied the algorithm above?

☐ Receive Credit for this lab guide

Submit the completed lab guide Google Classroom to receive credit

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